Case 23: Spatially varying fuel quadrants

1 Verification purpose

This case verifies raster lookup and fire-behavior coupling when a propagating front crosses four fuel-model regions. Flat terrain and zero wind isolate fuel identity. The verification question is whether every quadrant's maximum spread rate is within 10 percent of its fuel-specific reference.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The level-set normal speed is evaluated from the fuel model at each active cell. Because the environmental forcing is uniform, changes in V_n across the domain should follow only changes in the fuel-model classification raster. Correct row orientation and quadrant masking are therefore part of the test.

At each active cell the fuel code selects a complete Rothermel parameter row, and the local no-wind/no-slope rate is

$$R_0(x,y) = \frac{I_R(x,y)\,\xi(x,y)}{\rho_b(x,y)\,\epsilon(x,y)\,Q_{ig}(x,y)}. \tag{1}$$

Thus the raster lookup changes the full nonlinear fuel response, not a post-hoc scalar multiplier. The formulation follows Andrews (2018); the Guide supplies the four BEHAVE reference values.

3 Derivation of expected behavior

The independent regional targets are 0.080, 0.472, 1.492, and 0.871 m min^{-1} for Fuel Models 8, 7, 4, and 2 in the northwest, northeast, southwest, and southeast quadrants, respectively. These values also predict a visibly noncircular front after it enters all four regions. The Guide historically reported ELMFIRE rates of 0.080, 0.551, 1.974, and 0.872 m/min for FB8, FB7, FB4, and FB2, noting discrepancies for FB7 and FB4. Those historical outputs are preserved here as context only; the generated metric table below is the current evidence and is never replaced by those values.

All four references use the no-wind/no-slope Rothermel equation $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ with $Q_{ig} = 250 + 1116M_f$ and the generated moisture vector $(0.06, 0.07, 0.08, 0.60, 0.90)$. Substitution of each generated standard fuel model gives

fuel model	I_R (kW m^{-2})	R_0 (m min^{-1})
8	175.566	0.079552
7	406.479	0.470579
4	2272.300	1.493256
2	669.483	0.871756

and hence the report references $0.080,\ 0.472,\ 1.492$, and 0.871 m min^{-1}, respectively. For each fuel model, $e_R = |R_{\text{max}} - R_{\text{ref}}|/R_{\text{ref}}$; the table substitutes the corresponding row above and the prescribed relative-error limit.

4 Verification metrics and acceptance criteria

The maximum finite spread rate in each quadrant is compared with its corresponding target. Regional maxima preserve the local head-fire response and avoid mixing physically distinct fuels. Every component uses 10 percent relative tolerance, and PASS requires all four components.

5 Simulation configuration and predicted outcome

Four equal quadrants on a 126×126, 30 m grid contain Fuel Models 8, 7, 4, and 2. Terrain is flat, wind is zero, moisture is uniform, and ignition is central. The fastest extension is predicted in Fuel Model 4 and the slowest in Fuel Model 8.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

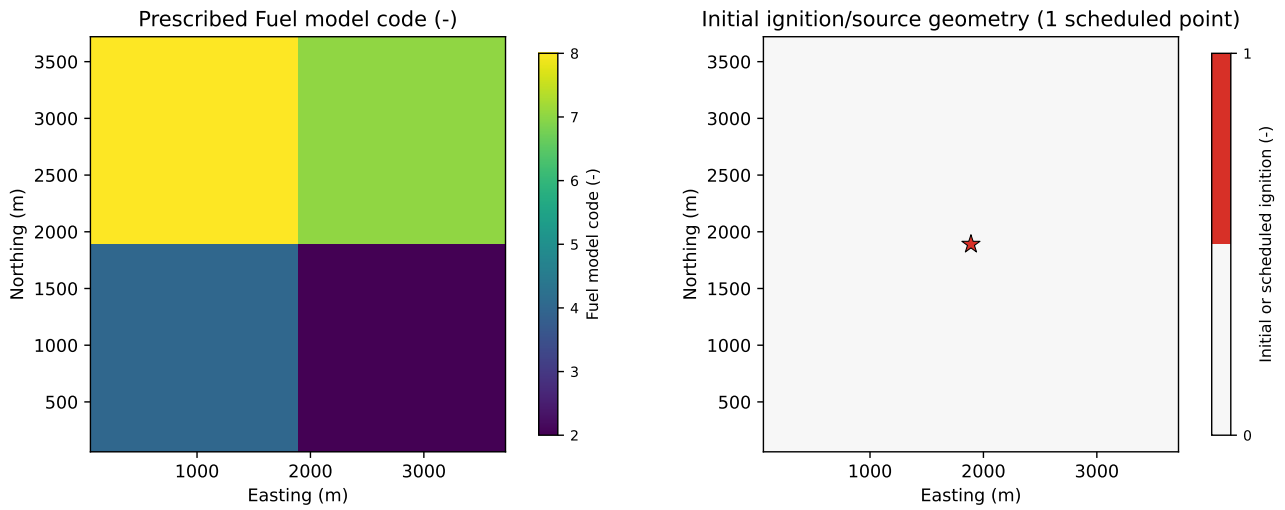


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

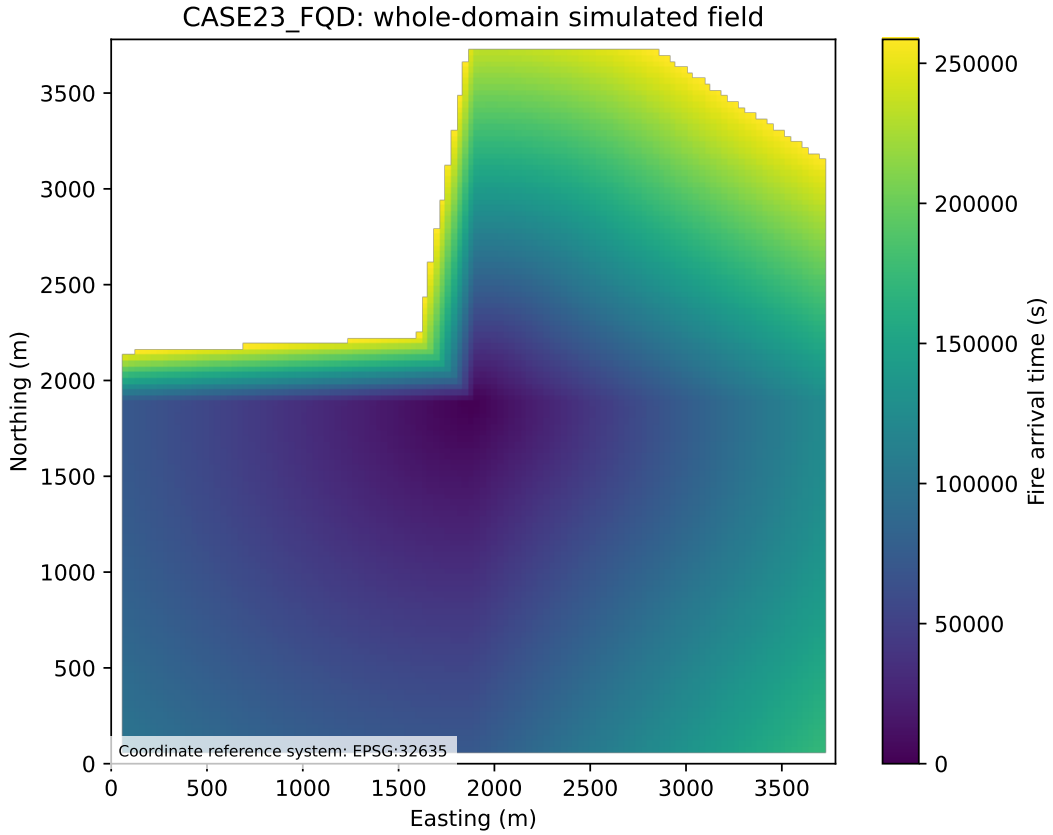


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / FB8 head ROS	0.08 m/min	0.0795524 m/min	PASS
main / FB7 head ROS	0.472 m/min	0.470579 m/min	PASS
main / FB4 head ROS	1.492 m/min	1.49326 m/min	PASS
main / FB2 head ROS	0.871 m/min	0.871756 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A quadrant swap indicates raster orientation or mask errors; a fuel-specific discrepancy indicates fuel-table selection or fire-behavior coupling. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test samples four fuel models under one moisture state and does not validate the empirical fuel parameters. This case provides implementation verification evidence only.